

Solution Requirements

Date	5 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Authentication	Users can sign up, log in, and log out securely using email and password. Passwords are hashed and sessions are managed securely.	High
2	Audio Upload & Management	Users can upload or record audio explanations. The system validates file type (mp3, wav, m4a) and size, and shows upload progress.	High
3	Concept Reference Selection	Users can search and select the concept/topic to evaluate their explanation against from a predefined concept library.	High
4	Audio Transcription	The system converts uploaded audio to text using a speech-to-text model with high accuracy.	High
5	Feature Extraction & Embedding	The system extracts meaningful features from the transcript and generates embeddings for semantic representation.	High
6	Evaluation & Scoring	The explanation is compared with reference content and scored on similarity, keyword match, completeness, and clarity using a scoring engine.	High
7	Results Visualization	Users can view detailed results including score breakdown, feedback, matched keywords, and waveform playback.	Medium
8	Report Generation & Export	Users can download the evaluation report as a PDF and share it.	Medium
9	User Dashboard & History	Users can view their past evaluations and track progress over time.	Medium
10	Notifications	The system provides real-time notifications for upload status, processing completion, and report readiness.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should provide fast response times for all user interactions and processing.	Page loads in < 2 seconds; Audio processed in < 30 seconds (for 5 min audio)
2	Scalability	The system should handle increasing number of users and audio files without performance degradation.	Supports up to 10,000 concurrent users and 1M+ audio files
3	Security & Data Privacy	User data and audio files must be encrypted in transit and at rest with proper access controls.	End-to-end AES-256 encryption; Role-based access control
4	Reliability & Availability	The system should be reliable and available to users with minimal downtime.	99.9% uptime per month; Auto-failover enabled
5	Usability & Accessibility	The system should be easy to use and accessible to all users, including those with disabilities.	Complies with WCAG 2.1 AA standards
6	Other (Maintainability)	The system should be easy to maintain and update with clean code and modular architecture.	Code coverage ≥ 80%; Modular, well-documented code